

Question ID: 7cab9fe1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	Medium

Question

Which expression represents the result of increasing a positive quantity w by **43%**?

Answer

- A. **1.43** w
- B. **0.57** w
- C. **43** w
- D. **0.43** w

Correct Answer: A

Rationale

Choice A is correct. The result of increasing a positive quantity w by $x\%$ can be represented by the expression $(1 + \frac{x}{100})w$. Therefore, the result of increasing a positive quantity w by **43%** can be found by substituting **43** for x in the expression $(1 + \frac{x}{100})w$, which gives $(1 + \frac{43}{100})w$, or **1.43** w . Thus, the expression **1.43** w represents the result of increasing a positive quantity w by **43%**.

Choice B is incorrect. This is the result of decreasing a positive quantity w by **43%**.

Choice C is incorrect. This is the result of increasing a positive quantity w by **4,200%**.

Choice D is incorrect. This is the result of decreasing a positive quantity w by **57%**.